Next Nanotechnology 1 (2023) 100003

Contents lists available at ScienceDirect

Next Nanotechnology

journal homepage: www.sciencedirect.com/journal/next-nanotechnology

Discussion

Organic light-emitting diodes, what's next?

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ARTICLE INFO

Keywords:
OLEDs
Display
Lighting
Stability
Cost per lumen

ABSTRACT

Organic light-emitting diode (OLED) technology has made significant advances in performance and becoming widely used in smartphone displays. However, the stability of efficient blue emitters, such as phosphorescent and thermally activated delayed fluorescent emitters, remains a challenge. While white OLEDs have the potential to revolutionize indoor lighting, their high cost per lumen due to efficiency roll-off and manufacturing costs is a limiting factor. To overcome these challenges, researchers must explore new material systems and device architectures to reduce efficiency roll-off and develop cost-effective film preparation processes. With continued increases in stability and decreases in cost, there is significant interest in scaling up OLED lighting and exploring its applications in other fields such as biomedical devices, anti-counterfeit labels, and optical communication.

Current status of OLEDs

An organic light-emitting diode (OLED) is a solid-state device that uses a stack of organic thin films sandwiched between an anode and a cathode. When holes and electrons are injected from the anode and cathode respectively, excitons, which are bound states of hole-electron pairs, can form in fluorescent or phosphorescent molecules/polymers in the organic layer(s). Electroluminescence can be achieved when the excitons relax from their excited states to ground states (Fig. 1). OLEDs are popular for their low energy consumption, light weight, and superior luminescence properties, and have been widely used in flat panel displays for smartphones, TV screens etc.

Since the first efficient OLED was developed in 1987, the field of OLED research and development has grown rapidly, and there have been significant improvements in both efficiency and lifetime through advancements in materials systems and device architectures. Various types of emission mechanisms have led to the development of efficient organic electroluminescent (EL) materials. It is widely recognized that fluorescent, phosphorescent, and thermally activated delayed fluorescent (TADF) emitters represent three generations of organic EL materials. In OLED devices, phosphorescent and TADF emitters have the ability to reach 100 % internal quantum efficiency (IQE) by fully utilizing both singlet and triplet excitons, while fluorescent can only use the singlet exciton and limited with an IQE of 25 %. As a result, most high-performance OLEDs are being developed using phosphorescent and TADF emitters, particularly those with high horizontal dipole ratios which can enhance light extraction out of the devices. Additionally, advancements in device architecture have also led to the development of complex stacks consisting of multiple layers with specific functions, such as carrier injection, transport, and blocking layers. These layers are designed to form and confine excitons in the emitting layer, for targeting near-unity recombination efficiency. Due to these efforts, phosphorescent and TADF OLEDs have achieved EQEs over 35 %, making OLED displays the most prevalent smartphone screens currently available.

What's the next for OLEDs?

However, despite these achievements, operational stabilities of blue phosphorescent and TADF emitters remain a challenge, which is why triplet-triplet annihilation (TTA)-based blue emitters are currently used in commercial OLED displays [1]. To further reduce energy consumption in commercial OLEDs, more efficient and durable blue phosphorescent and TADF emitters or emitters with other types of emission mechanisms, such as hyper-fluorescence, are being actively explored. One way to improve the stability of efficient blue emitters is by reducing the probability of non-radiative events by shortening the radiative lifetime of excitons. By matching the emitters with the resonant modes of an optical cavity, the rate of spontaneous emission can be increased [2], which is another way for enhancing the performance of phosphorescent and TADF emitters in addition to molecular design.

Despite the widespread use of OLED displays in smartphones, OLED lighting products are still not as common. White OLEDs (WOLEDs) have the potential to revolutionize indoor lighting with their soft, natural, and non-

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https://doi.org/10.1016/j.nxnano.2023.100003
Received 6 April 2023; Accepted 9 April 2023

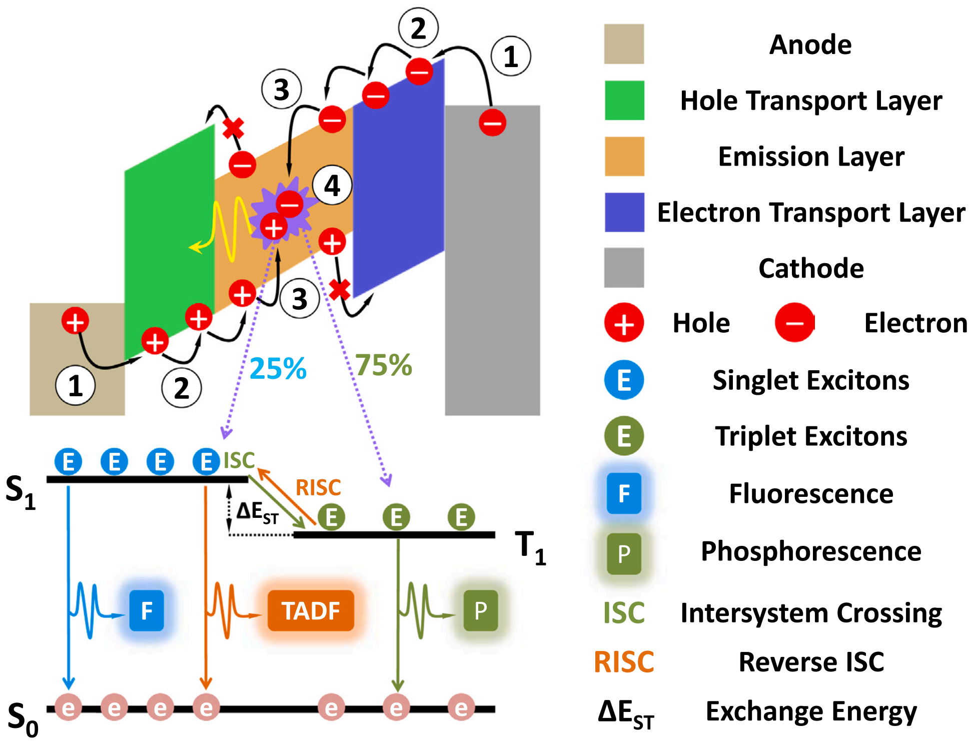

Fig. 1. Working mechanism of OLEDs. The arrows labeled 1–4 represent the steps of carrier injection, carrier transport, exciton generation, and radiative decay of excitons, respectively.

glare light. Lab-created WOLEDs have been shown to have power efficiencies above 110 lm/W [3], which exceeds traditional incandescent bulbs and fluorescent lamps with efficiencies of < 15 and 60–90 lm/W respectively. This would greatly reduce carbon emissions as indoor lighting accounts for approximately 15 % of worldwide electrical energy consumption [4]. One major challenge in using OLEDs for lighting is the stability of blue emitters at practical brightness levels. TTA blue emitters are not efficient enough for reducing power consumption for lighting applications. To address this, researchers are developing more efficient and durable blue phosphorescent and TADF emitters. Another potential solution is to use electroluminescent quantum dot emitters, which have the potential to be more stable and could be used as an alternative for blue emitters in OLED lighting applications. On the other hand, another obstacle for commercializing OLED lighting is that its cost per lumen (over $50 per kilolumens) is still much higher than existing lighting technologies. The amount of light an OLED produces, measured in lumens, is determined by its luminance (measured in candela per square meter) and the size of the emitting area. To maximize the amount of light emitted from a fixed light-emitting area and reduce the cost per lumen of OLED lighting products, it is desirable to operate OLEDs at high luminance levels (greater than 5000 cd/m^2) [5]. While much effort has been put into achieving the theoretical maximum efficiency of OLEDs, high-performance OLEDs still struggle with maintaining efficiency and stability at high luminance levels.

The efficiency roll-off issue in OLEDs, which causes a reduction of efficiency at high luminance, is a complex and nonlinear problem related to the intrinsic properties of organic semiconductors, which are not fully understood in many cases. It is generally believed that the efficiency roll-off is affected by bimolecular processes, such as exciton-exciton and exciton-polaron annihilations. Additionally, the difference in electron and hole mobilities of most organic semiconductors also contributes to the efficiency roll-off [6]. Imbalances in transport properties can lead to issues such as current leakage or an accumulation of majority carriers, which can negatively impact exciton generation efficiency and cause significant exciton-polaron annihilation at high luminance levels. To overcome these issues, it is crucial for researchers to explore for new materials systems and device architectures that can substantially reduce roll-off. On the other hand, the weak interaction between organic semiconductor molecules leads to a relatively low current density window for OLEDs (around 100 mA/cm^2), which is about three orders of magnitude lower than those (around 100 A/cm^2) in commercial inorganic LEDs. This means that, even assuming the same efficiency, to achieve the same lumen output as an inorganic LED chip, an OLED panel with a much larger area is required. The current density window of OLEDs is typically limited by resistance and/or permittivity of a particular "bottle neck" layer in the organic stack. Under the dynamic equilibrium of current flow, electric field distribution in the organic stack is highly uneven, resulting in a potential difference across the "bottle neck" layer that can be several orders of magnitude higher than that across the other layers [7]. The "bottle neck" layer would be broken down first before the other layers. To increase the current density window of OLEDs, it is necessary to enhance the dopant and host molecule mobilities in the emissive layer and reduce net polarization charges at organic heterointerfaces.

Another essential requirement for commercializing OLED lighting is to decrease its cost. This can be accomplished by using solution processes which are of much lower cost comparing to vacuum-based processes. All solution-processed OLEDs have been achieved by using orthogonal solvent systems [8]. However, using orthogonal solvent systems alone may not be enough to fully transfer the high performance of OLEDs to solution-processed OLEDs, as many advanced small-molecular organic semiconductors have similar polarities. To address this issue, new methods for creating anti-solvent organic semiconductor films using chemical crosslinking via UV or thermal-triggered reactions need to be developed. This type of crosslinking can specifically target the nonconjugated side chains without breaking the conjugation in the backbone, which helps to preserve the charge transport properties of the organic semiconductor films [9]. To make further progress in the development of cost-effective OLEDs, it is crucial to focus on developing liquid organic semiconductors by incorporating long-chain alkyl groups and crosslinkable functional groups. Once these liquid organic semiconductors are developed, it will be possible to produce organic semiconductor films at a low cost without using vacuum or solvent, through

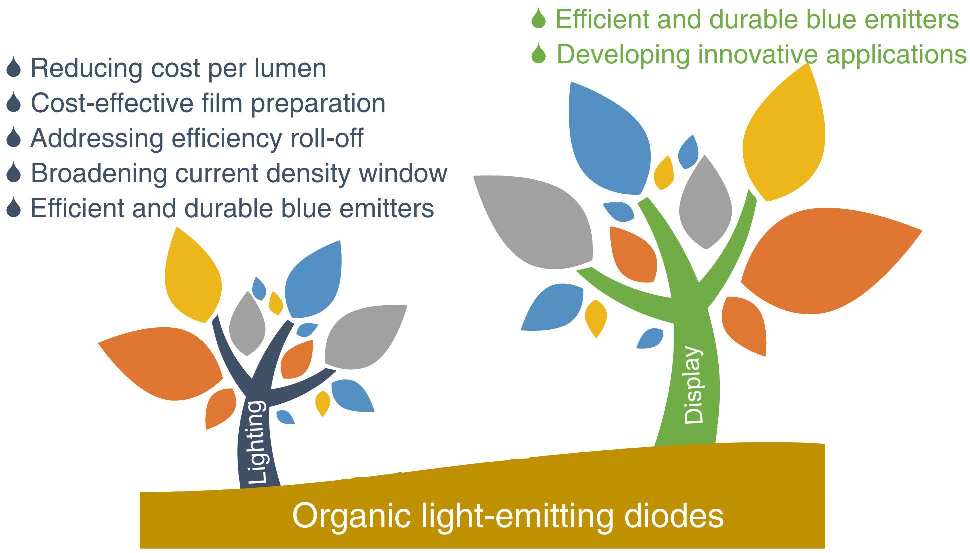

Fig. 2. Next-stage research focus for OLEDs.

the process of curing liquid precursors of organic semiconductors using UV or thermal curing methods, similar to the production of adhesives.

In addition to display and lighting applications, OLEDs are also being used as light sources for various applications such as biomedical devices, anti-counterfeit labels [10,11], optical communication and more. The flexibility of OLEDs allows them to be fabricated onto various substrates, enabling the development of compact, lab-on-chip applications by integrating them with other optical and electronic components. However, the performance of OLEDs is still inferior compared to that of semiconductor LEDs and laser diodes. These OLED applications are suitable for situations that prioritize compactness over high performance. As researchers continue to focus on improving OLED stability and reducing manufacturing costs (Fig. 2), it will be interesting to see if the industry can achieve scaled production of OLED lighting soon and further expand their potential applications in various fields.

CRediT authorship contribution statement

All authors have made contributions to this editorial and approved for its publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (62174067, 62175085, 61905086), Science and Technology Development Planning of Jilin Province (Project No. 20200201296JC, 20230101061JC) and grants from the Research Grants Council of the Hong Kong Special Administrative Region, China (Project Nos. 11300418 and 11300419).

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